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Evaluating Machine Learning Models for Sentiment Analysis of YouTube Comments and Creating an Accessible Web Application for Comment Analysis

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Abstract

The popularity of the YouTube platform for content creation and its rapid accessibility to users is growing significantly. YouTube is eager to receive feedback on the material they have produced and made available to the public. Most research in this field has focused on analyzing static data rather than real-time streaming data. This study provides comprehensive investigation of machine learning-based sentiment analysis of live YouTube comments. This research paper proposes the development of an effective sentiment analysis model with user friendly web application that can accurately classify live YouTube comments into three groups: neutral, negative, and positive. The machine learning models, like extreme gradient boosting, random forest, logistic regression, adaptive boosting, gradient boosted decision tree, and bagging classifier along with natural language processing techniques have been used to implement the proposed system. In this research work, we used the Kaggle dataset to train the machine learning models and evaluated the performance. performance of these models using performance metrics like accuracy, precision, recall, and f1 score. The experimental work showed that the bagging classifier with 95.8 % accuracy performed well as compared to others. Finally, the bagging classifier was deployed to create a user-friendly web interface to make the developed system an effective YouTube comment analyzer for live and online video. Hence, by using this system, YouTube content creators can quickly analyze the impact of their videos on society.

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1. Introduction

YouTube comment analysis is a technique to learn about the comments (positive, negative, or neutral) received on publicly available YouTube content [1]. The comment analysis is essential to provide feedback on publicly available YouTube video content. This technique is nowadays becoming very important to identify trends and people's likes and dislikes. It is also important to identify in which situations people remain neutral. Due to the huge amount of information and wide variety of comments to do the comment analysis manually, is very challenging. The comment classification can be implemented using the Machine Learning (ML) and Deep Learning (DL) [2], [3]. Also with these ML and DL models for handling linguistic feature extraction Natural Language Processing (NLP) is also used [4].

Selecting one out of the available YouTube comment analysis techniques is one of the issues content creators are facing. So to resolve these issues, this research work planned to do experimental work and suggest the one who will be better suited for the YouTube comment analysis job. And it has been found that some scholarly articles those use ML techniques such as Naive Bayes (NB), support vector machines (SVM), logistic regression (LR), Decision Tree (DT), Maximum Entropy (ME), K-Nearest Neighbors (KNN), and some semi-supervised learning algorithms[5].

The experimental research work that is presented in this paper aims to achieve the following objectives:

1.1. Objectives

- To develop an effective YouTube comment sentiment analysis system using NLP techniques and ML models.
- To deploy the developed model and implement the user friendly WebApp for analyzing live YouTube content.

Followed by this introduction section, the paper is organized in the following sections, such as related work, proposed method, experimental work, and performance evaluation. The results and discussion section presents the results obtained from the experimental work, while the final portion contains the concluding remarks and the future scope of the work.

2. Related work

This section presents the theoretical study related to the chosen research domain and provides insight into different relevant domains concerning the research questions raised in the previous section. The scholarly Articles in this research domain from various top publishers have been studied. Based on the literature study and review research gaps have been identified. ML-based analysis of YouTube comments is an expanding field of study with applications in user engagement, sentiment analysis, and hate speech detection. Key Works in this field will be covered in this overview of the literature.

Applying comment analysis to generated content is on the rise due to the popularity of Internet video platforms like YouTube. Hence, analyzing the comments can provide valuable insights into user sentiment [6]. Recent research has explored the application of ML algorithms to analyze YouTube comments and their influence on the impact of videos on viewers [7]. The main aim of the [8] study is to provide the full concept of sentiment analysis techniques. A survey article [5] presents a brief survey of techniques to analyze opinions, posted by users about a particular video. The review article [6] presents techniques to identify comment's tone such positive, negative and neutral. A textual based review using fuzzy classification model is proposed in [9] that uses the movie review dataset for evaluating the performance of the developed model. The system proposed in [10] uses six different ML algorithms, including RF, LR, SVM, DT, KNN, and NB. SVM and NB are combined for better accuracy in [11], and it has been found that the combination produces a better accuracy level and stronger performance. A sentiment analysis approach to user comments using NLP is proposed in [12]. The research work presented in [13] uses well known statistical measures and ML models to perform sentiment analysis. The authors emphasize the difficulty presented by the informal nature of YouTube comments. That frequently include grammatical errors and slang. Web scraping is a technique for obtaining data from websites by means of HTML tags. It is possible to obtain user remarks from the comment section with web scraping. This technique is implemented in [14]. The following subsection puts more insights on the existing system by reviewing the literature to find the research gaps.

2.1. Literature review

The literature review presented in this subsection discusses recent research in this domain and gaps identified on YouTube comment analysis using ML techniques. Table 1 shows state-of-the-art literature review. From this Table 1, it has been found that some research gaps, such as the need for a more accurate and fast ML models, are required. Also, it is identified that a user-friendly web or mobile application is essential for ML-based YouTube comment analysis effectively.

Table 1. A state-of-the-art literature review on YouTube comments analysis using ML models.

Authors (Year)	Algorithms/Methodologies	Results	Key Findings and Future Scope
Musleh et al. [15] (2024)	Naïve Bayes	94.62%	Performances of six ML approaches compared and presented RF as better.
Maw et al. [3] (2024)	Long Short-Term Memory (LSTM), Transformer Network, and Gated Recurrent Unit (GRU)	Acc. 97% precision 0.97	LSTM proposed as better.
Uddin et al. [16] (2023)	Attention model	Acc. 92.8%	Useful guidelines for sentiment analysis on several languages in the future.
K.M. Kavitha et al. [1] (2020)	Bag of Words and Association List	Accuracy, Precision and recalls are evaluated for multilevel classification.	Automatic YouTube comments analysis tool is proposed. Multilingual phrases or words
Cunha et al. [17] (2019)	Convolution Neural Network	Acc. 60%–84%	Better accuracy than the statistical models.

3. Dataset

Since the study conducted for this paper is predicated on ML and NLP concepts, we must take into account the material, that is, the dataset—that was used to conduct the experiments. The dataset [18] used to train the ML models is publicly available at Kaggle. The distribution of the dataset is shown in the Fig. 1. One observation from the collected YouTube data set is that the dataset is imbalanced in nature (uneven distribution of positive, negative, and neutral comments). A value of 0 represents negative sentiment, while values of 1 and 2 represent neutral and positive sentiments, respectively. The dataset used for the experimental work is made up of YouTube comments that were scraped automatically using Google's application programming interface (API) Key. The users must use the universal resource locator (URL) in the text's input field of the *Streamlit* application. This serves as the basis for retrieving comments from the specified YouTube video. The data frame is used to save the extracted text of the comments and the associated sentiment labels.

4. Proposed System

In this section, the proposed system, as depicted in Fig. 2, is introduced to address the identified research gaps. The following subsections will provide detailed explanations of the proposed methods. Also, described various steps adopted to implement it in detail. Next, ML algorithms are used to classify the comments into several categories. The cleansed data from the pre-processing is utilized to train the ML

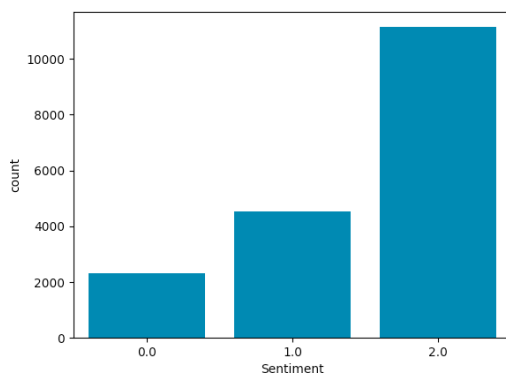


Fig. 1. Distribution of number of positive, neutral and negative samples in dataset

models. The optimized and deployed trained model with favorable precision and high accuracy is used to develop Web applications. The deployed web application gathers data from the streaming YouTube video and divides it into three groups, like negative, neutral, and positive.

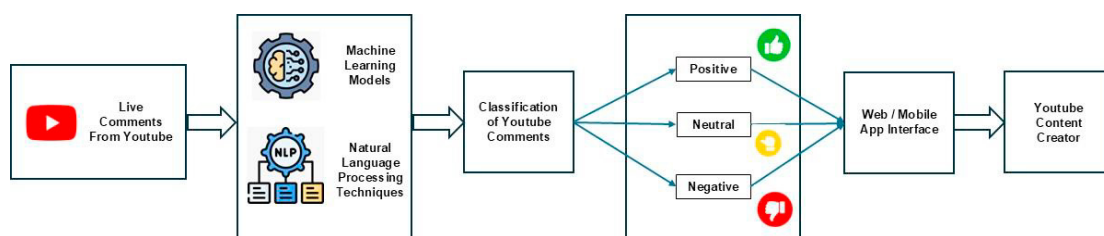


Fig. 2. Proposed system for Youtube Comment Sentiment Analysis ML models and NLP techniques

4.1. Data Preprocessing using NLP techniques

The comments are standardized through data preprocessing before conducting sentiment analysis [19]. The following subsection discusses some basic steps in NLP.

4.1.1. Text processing

It is one of the methods used in NLP to convert raw text data into machine understandable format. In The first step is "tokenization," i.e., splitting long sentences into phrases or words. The subsequent text in this process is the "lowercasing". Followed by lower casing the important step in NLP "stopword removal" is applied. "Text cleaning" is also required to remove unwanted elements such as punctuation, special characters and numbers that may clutter the analysis.

4.2. Feature extraction

The feature extraction in NLP is to convert raw text data into numerical form so that machine can interpret and analyze it easily. The "term frequency-inverse document frequency TF-IDF" and "Bag-of-Words", and more advanced word embedding techniques like "Word2Vec" are used for feature extraction.

4.2.1. Text analysis

Text analysis assesses the text's emotional tone and categorizes it as positive, negative, or neutral. The Natural Language Toolkit (NLTK) is a collection of English-language modules and applications used for tokenization, semantic reasoning, stemming, tagging, text classification, and parsing. The preprocessed data is then used for ML model training.

4.3. ML models

The ML models adopted for experimental work are extreme gradient boosting (XGB), random forest (RF), logistic regression (LR), gradient boosted decision tree (GDBT), AdaBoost (AdaB), and the bagging classifier (BgC). The XGB [20] is ensemble technique of classifier in which weak learners are combined together. Some of the features of this technique are part of speech, TF-IDF, "Bag of Words", N-gram, and Opinion Lexicon [10]. For sentiment analysis, RF can be useful because it maintains balance between accuracy and interpretability [21]. AdaB [22] is the one more ensemble classifier that updates its weight according to their output. According to [23], for the binary sentiment analysis, the LR model is more suitable. The BgC is used in sentiment analysis to increase the robustness and accuracy of sentiment classifiers [24] by preventing overfitting. One important component of the YouTube comment sentiment analysis project is a web application that lets users enter video URLs and starts sentiment analysis on related comments. It displays the data in a tabular format that is simple to read, along with the sentiment distribution and changes over time.

5. Web Application

To demonstrate the working of the deployed ML model using a BgC classifier, a user-friendly web application is developed that takes input as YouTube live comment as shown in Fig. 3. And implements the developed ML model to analyze comments into three categories as shown in the Figures (positive comments Fig. 5, negative comments Fig. 6, and Neutral comments Fig. 7). Fig. 3 is the user interface (UI). Through

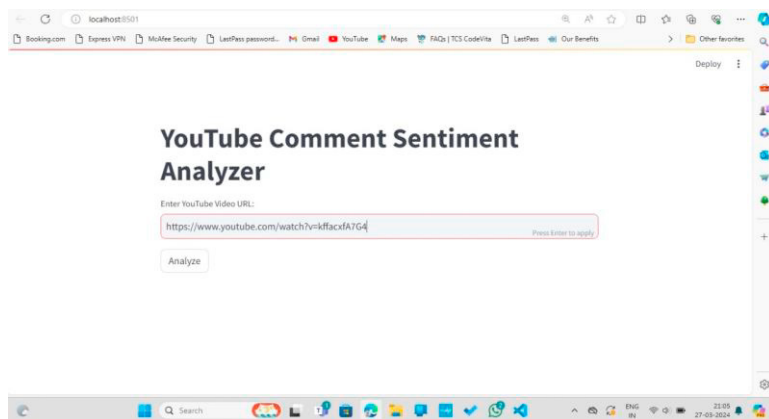


Fig. 3. User interface

this, users can directly interact with the system and provide input in the form of a YouTube video URL.

5.1. Experimental work

The experimental work was carried out utilizing the computing platforms and programming languages like Python and NLTK libraries. Sentiment analysis on YouTube comments can be done with a range of frameworks and packages that offer APIs for NLP tasks. Also some advanced tools like "TextBlob" and "Valence aware dictionary and sentiment reasoner (VADER)" are used in the work. Performance analysis of experimental work is shown in the following section.

6. Performance metrics

For the evaluation of the ML learning classifier model, the most commonly used performance metrics are defined, using the confusion matrix. The terminologies used to define the performance metrics are as follows:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

The accuracy and precision performance parameters have been used to evaluate the performance of the implemented model. The accuracy gives how the model is classifying the user comments into three different categories like negative, neutral, and positive.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

The precision parameter is used to measure the performance of the model to predict the positive predictions. It is represented mathematically, as shown in Equation 2.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

The recall is one of the metrics to evaluate the ML classifier model. It is used to identify the amount of FN a classifier produces.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$f_1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

Using these performance metrics developed model's performance is evaluated and presented in the following results and discussion section.

7. Results and Discussion

Table 2. Performance evaluation of developed model

Models	Accuracy	Precision	Recall	f1-score
XGBoost	0.9221	0.9304	0.7230	0.6152
Random Forest	0.9308	0.9361	0.5211	0.5211
Logistic Regression	0.9351	0.9661	0.59	0.61
Gradient Boosted Decision Tree	0.9394	0.9798	0.8443	0.8021
AdaBoost	0.9510	0.9596	0.7520	0.7909
Bagging Classifier	0.9582	0.9370	0.9190	0.8802

A Google Colab online editor and compiler is used to perform the experiments. The ML model adopted to develop the proposed system includes XGB, RF, LR, GDBT, AdaB, and BgC along with NLP text pre-processing techniques. The models are trained, and their performances are evaluated using accuracy and precision as performance measures. The training performances are shown in Table 2. It displays the precision and accuracy of each model at the time of experimentation. From Table 2, it can be observed that the bagging classifier outperforms the other classifiers in terms of accuracy with 95.82%. And, additionally,

the Table 2 shows that the gradient-boosted decision tree performs better in terms of precision with 97.98%. Fig. 4 shows the performance evaluation of algorithms that train the proposed system using accuracy and precision as performance metrics. It has been observed that BgC is performing well in terms of accuracy compared to others. In the case of precision, the GBDT is the best among others.

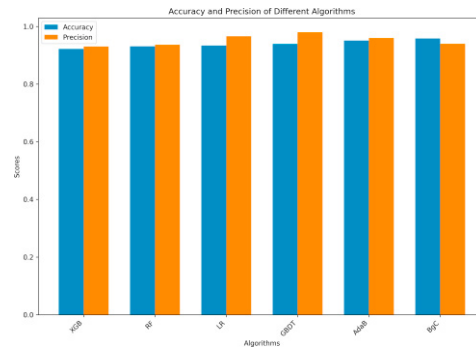


Fig. 4. Performance evaluation of algorithms used to train the model with accuracy and precision as performance metrics.

7.1. Performance analysis

Table 3. Performance analysis of proposed method with state-of-art techniques

Methods/ Performance metrics	Musleh et al. [15]	Maw et al. [3]	Uddin et al. [16]	K.M. Kavitha et al. [1]	Cunha et al. [17]	Proposed Method
Accuracy	94.62%	97%	92.88%	91.3%	60%–84%	95.82%

Table 3 shows the comparative analysis of developed model with existing system proposed by [15], [3], [16] and [1]. It has been observed that the proposed system is performing better as compared to existing systems considered here for the comparison.

7.2. Working of the developed Web App

Positive Comments

0 :
 ""Thank you for providing us with valuable concepts on YouTube. Your content has been insightful and enriching, and I truly appreciate the effort you put into sharing all the delightful authentic goan food and the places."
 1 :
 "Thank you for the Kuttikars video at Old Goa. I've been back to Goa and got a few friends touring the state wanting to explore goan cuisine. I needed a place for sea food only. The fish thali here at Vidas also looks tempting. There are a few restaurants near the river at Asolna. Joe's river cove is located towards Assolna Cavelossim road and Star light in Tariwado-Asolna. I'd recommend star light for lunch as their quality gets compromised with the evening crowd. Don't forget to try the stuffed mushrooms at Joe's river cove if you do visit and beef chilly at star light."

Fig. 5. Positive Comments

The comments captured through the Google API are the input to ML model, that classify these comments into three different categories, like positive neutral and negative. The following figures display the corresponding results of this developed system. When we applied comment analysis to the input comments,

```
23 :
  "Very upscale. It is very nice inside. I like the large windows. I would have
  expected it to be busy? I would expect the food to be more refined. I imagine
  the chef is a real chef."
]
```

Negative Comments

```
▼ [
  0 : "Look at the rates 🍷"
  1 :
    "Lesley, On the Bharit, i think its the black crushed Nori made of seaweed and
    is used in Sushi"
]
```

Fig. 6. Negative Comments

the results are shown in Fig. 5 as positive comments. And the result as unfavorable remarks is shown in Fig. 6. As seen in Fig.7, this approach also separates the neutral remarks from the overall comments and labels them as negative comments.

```
]
```

Neutral Comments

```
▼ [
  0 :
    "Costs we paid: brinjal bhart on chapati Rs 220, tambdi bhaji samosa Rs 220,
    assortment of phodis Rs 220, Goan fish thali Rs 320, kingfish rawa fry Rs 600,
    prawns masala fry Rs 490, beetroot, pineapple, pomegranate salad Rs 200,
    khadatli tora Rs 50, teesrio sukha Rs 280, veg Kolkapuri Rs 280, coco loco Rs
    380, serradura Rs 200m kangachyo nevryo Rs 200"
  1 : "Try doria dege at old goa.... fish thali ..."
  2 : "Why can't you try Mahesh Bar and restaurant is Ramnathi, Ponda, Goa?"
  3 :
```

Fig. 7. Neutral Comments

8. Conclusion

In this research work, we identified the potential scope for YouTube comment analysis by reviewing the state of art literature. After conducting experiments and research, we also determined which ML technique was more effective for effectively analyzing the comments on YouTube. For our experimental work, we selected the bagging classifiers. With an accuracy of 95.88%, we discovered that the bagging classifier was outperforming the others. We have created a user-friendly web interface in order to make system an effective

YouTube comment analyzer for live and online video. Any YouTube content creator can quickly analyze the impact of their videos on society by using this web interface.

One of the future directions in this research domain is to use ensemble learning techniques, which can also be considered for implementing comment analysis in multilingual contexts. Additionally, strategies to handle imbalanced datasets could be a focus of future research in this area.

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